Copilot

Sure! Let's break down the recursion process step-by-step for the input `"abc"`:

Initial Call

The `main` method calls `printCombinations("abc", 0, "")`.

First Level of Recursion

- `index = 0`, `Combination = ""`
- Loop through each character in `"abc"`:
  - For `i = 0`: Call `printCombinations("abc", 1, "a")`
  - For `i = 1`: Call `printCombinations("abc", 1, "b")`
  - For `i = 2`: Call `printCombinations("abc", 1, "c")`

Second Level of Recursion

For the call `printCombinations("abc", 1, "a")`:

- `index = 1`, `Combination = "a"`
- Loop through each character in `"abc"`:
  - For `i = 0`: Call `printCombinations("abc", 2, "aa")`
  - For `i = 1`: Call `printCombinations("abc", 2, "ab")`
  - For `i = 2`: Call `printCombinations("abc", 2, "ac")`

Third Level of Recursion

For the call `printCombinations("abc", 2, "aa")`:

- `index = 2`, `Combination = "aa"`
- Loop through each character in `"abc"`:
  - For `i = 0`: Call `printCombinations("abc", 3, "aaa")`
  - For `i = 1`: Call `printCombinations("abc", 3, "aab")`
  - For `i = 2`: Call `printCombinations("abc", 3, "aac")`

Base Case

For the call `printCombinations("abc", 3, "aaa")`:

- `index = 3`, which equals the length of `"abc"`.
- Print `Combination = "aaa"`.

This process repeats for each combination, generating and printing all possible combinations of the characters in `"abc"`.

Visualization

Here's a simplified visualization of the recursion tree for the first few levels:

```
printCombinations("abc", 0, "")
|
|-- printCombinations("abc", 1, "a")
|   |-- printCombinations("abc", 2, "aa")
|   |   |-- printCombinations("abc", 3, "aaa") -> prints "aaa"
```

```
|   |   |-- printCombinations("abc", 3, "aab") -> prints "aab"
|   |   |-- printCombinations("abc", 3, "aac") -> prints "aac"
|   |
|   |-- printCombinations("abc", 2, "ab")
|   |   |-- printCombinations("abc", 3, "aba") -> prints "aba"
|   |   |-- printCombinations("abc", 3, "abb") -> prints "abb"
|   |   |-- printCombinations("abc", 3, "abc") -> prints "abc"
|   |
|   |-- printCombinations("abc", 2, "ac")
|       |-- printCombinations("abc", 3, "aca") -> prints "aca"
|       |-- printCombinations("abc", 3, "acb") -> prints "acb"
|       |-- printCombinations("abc", 3, "acc") -> prints "acc"
|
|-- printCombinations("abc", 1, "b")
|   ...
|
|-- printCombinations("abc", 1, "c")
    ...
```

This continues until all combinations are printed. Let me know if you have any questions or need further clarification!